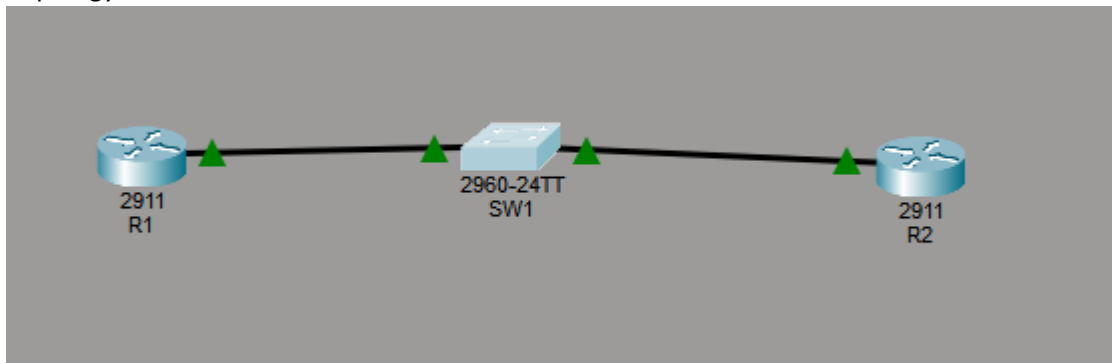


NET512 week 5 PRACTICAL screenshot:

1.) Topology:



2.) SW1 configuration:

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#z
Switch(config)#^
% Invalid input detected at '^' marker.

Switch(config)#^Z
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#CONFIGURE TERMINAL
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#INTERFACE VLAN 1
Switch(config-if)#IP ADDRESS 10.10.10.10 255.255.255.0
Switch(config-if)#NO SHUTDOWN

Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch(config-if)#EXIT
Switch(config)#
```

Copy

Paste

3.) Task 35 screenshot:

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#description connected to R1 fastethernet 0/0
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

4.)

```
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/2
Switch(config-if)#description connected to R2 fastethernet 0/0
Switch(config-if)#eint
^
% Invalid input detected at '^' marker.

Switch(config-if)#exit
Switch(config)#
```

Copy

Paste

5.) IP INTERFACE BRIEF:

```
Switch#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/1          unassigned      YES manual up          up
FastEthernet0/2          unassigned      YES manual up          up
FastEthernet0/3          unassigned      YES manual down        down
FastEthernet0/4          unassigned      YES manual down        down
FastEthernet0/5          unassigned      YES manual down        down
FastEthernet0/6          unassigned      YES manual down        down
FastEthernet0/7          unassigned      YES manual down        down
FastEthernet0/8          unassigned      YES manual down        down
FastEthernet0/9          unassigned      YES manual down        down
FastEthernet0/10         unassigned      YES manual down        down
FastEthernet0/11         unassigned      YES manual down        down
FastEthernet0/12         unassigned      YES manual down        down
FastEthernet0/13         unassigned      YES manual down        down
FastEthernet0/14         unassigned      YES manual down        down
FastEthernet0/15         unassigned      YES manual down        down
FastEthernet0/16         unassigned      YES manual down        down
FastEthernet0/17         unassigned      YES manual down        down
FastEthernet0/18         unassigned      YES manual down        down
FastEthernet0/19         unassigned      YES manual down        down
FastEthernet0/20         unassigned      YES manual down        down
FastEthernet0/21         unassigned      YES manual down        down
--More--
```

Copy

Paste

6.) Screenshot of shutting interface backup:

```
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastethernet 0/2
Switch(config-if)#shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

Switch(config-if)#end
Switch#
```

```
Switch#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	administratively down	down
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down

--More--

Copy Paste

Top

7.) Screenshot of interface bringing up:

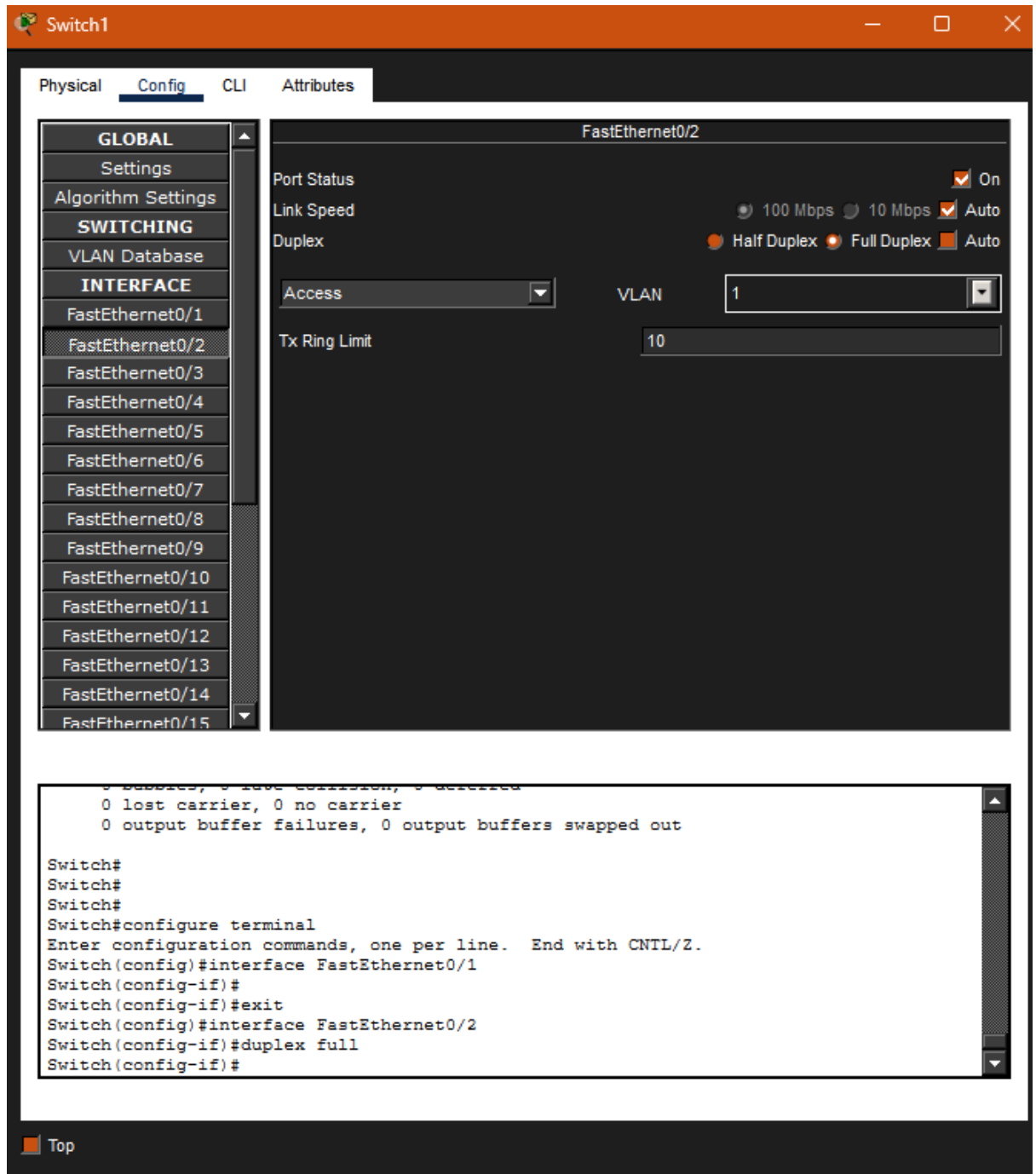
```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastethernet 0/2
Switch(config-if)#no shutdown
```

8.) Task 43:

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastethernet 0/2
Switch(config-if)#no shutdown
```

```
Switch#show interfaces fastethernet 0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.c96b.b302 (bia 0001.c96b.b302)
  Description: connected to R2 fastethernet 0/0
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
..
```

9.) Task 44:



Switch1

Physical Config CLI Attributes

GLOBAL

Settings

Algorithm Settings

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/1

FastEthernet0/2

FastEthernet0/3

FastEthernet0/4

FastEthernet0/5

FastEthernet0/6

FastEthernet0/7

FastEthernet0/8

FastEthernet0/9

FastEthernet0/10

FastEthernet0/11

FastEthernet0/12

FastEthernet0/13

FastEthernet0/14

FastEthernet0/15

FastEthernet0/2

Port Status ☒ On

Link Speed 100 Mbps 10 Mbps ☒ Auto

Duplex ☒ Half Duplex ☒ Full Duplex ☐ Auto

Access VLAN 1

Tx Ring Limit 10

0 collisions, 0 late collisions, 0 deferred
0 lost carrier, 0 no carrier
0 output buffer failures, 0 output buffers swapped out

Switch#
Switch#
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#duplex full
Switch(config-if)#

Top

10.)Task 45:

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/1
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#interface FastEthernet0/2
Switch(config-if)#duplex full
Switch(config-if)#exit
Switch(config)#interface fastethernet 0/2
Switch(config-if)#speed 10
Switch(config-if)#end
```

11.)Task 46:

```
Router>show ip interface brief
Interface          IP-Address      OK? Method Status        Protocol
FastEthernet0/0    10.10.10.2      YES manual up            up
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
Router>
```

```
Router>
Router>show interface fastethernet 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0040.0b29.8901 (bia 0040.0b29.8901)
  Internet address is 10.10.10.2/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
      Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
```

Reflection Question Answer:

47: when I set duplex mismatch between SW1 and R2 the symptoms I observed was that the interface status remained up/up and packet loss occurred.